

Archana P Patkar

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SUMMARY

Visionary network and security Engineer with over 8 years of experience in infrastructure design, assessment, deployment and maintenance. Expertise in threat detection, incident response, and security policy development. Committed to staying ahead of emerging cyber threats and ensuring robust protection for critical network assets. Passionate about delivering high-performance, secure, and scalable network and security solutions to meet evolving business needs.

KEY ACHIEVEMENTS

Firewall Migration Leader

Led firewall migration projects, reducing downtime by 30% across seven major data centers.

Security Enhancement

Enhanced threat mitigation efficiency by 25% through advanced security incident response workflows.

FedEx Site Configuration Success

Successfully configured 800 FedEx sites improving site connectivity by 40% in a year.

TECHNICAL STRENGTHS

Devices : Cisco switches 2960, 9300, 4K series, Cisco routers 4331, 4351, Siemens, Hirschmann switches, Cisco wireless access points, Juniper firewalls SRX3600, 4200, 4600, F5 firewalls, Palo Alto firewalls.

Protocols : EIGRP, OSPF, BGP, HSRP, VRRP, DHCP, SNMP, FTP, SSL, HTTP, DHCP, ICMP, DNS, ARP, IPv4, IPv6, SSH, Telnet, STP, MPLS, NAT, NTP.

LAN : TCP/IP, UDP, VLAN, LACP, subnetting, segmentation, virtualization, NFV, network automation using Ansible, Python3, load balancers, and proxy servers.

AAA/ Security: RADIUS, ACL, IPS/IDS, NAT, BPDU Guard, Root Guard, BIG-IP, Cisco ASA, Active directory, Radware Defense Pro/ APSolute vision, Gigamon.

Tools : Wireshark, SolarWinds, Packet Tracer, SDN, Infoblox, ISE, iPerf3, tcpdump, Wireless Controller, GNS3, Netflow, HPNA, SevOne, NSD, IFP, NETSCOUT, FireMon, NFVD, VNF, Atlas.

Operating systems : Windows 7/10, Linux, Cisco IOS, Junos OS, VCP.

WORK EXPERIENCE

Network Security Engineer III, Verizon, Branchburg NJ

May 2022- Nov 2024

- Conducted comprehensive network infrastructure assessments and designed, planned, and implemented multi-vendor security infrastructure using load balancers, firewalls, routers, and switches to ensure network stability, performance and 100% network availability.
- Led firewall migration and bandwidth augmentation projects, performed L1 tests, configured devices using Ansible, and creating detailed documentation including MOP/SOW/cabling sheets, policy changes, collaborating with cross-functional teams and key stakeholders utilizing network tools like INAM, SevOne, and IFP reducing the cost of operation by 80%.
- Tested new devices with IXIA to keep with risk profile. Monitored network diagnostics, mitigated threats, and responded to security incidents using tools like Radware DefensePro and Gigamon, reducing network threat impact by 70%.
- Expanded routing engines on virtual firewalls and Palo Alto, Fortinet, Juniper, F5 firewalls, including aspects of Active Directory integration and IAM for enhanced policy management, implementation, and configuration of network devices in complex environments, review of technical designs documentation.

Network Engineer, Princeton University, Princeton NJ

Nov 2021-Feb 2022

- Spearheaded the Aruba to Cisco migration project, designing an innovative process that significantly reduced downtime on the University network by 80%.
- Conducted wireless site surveys, diagnosed and resolved wireless troubleshooting across the University campus, working with Aruba access points, NetAtlas, and Airwave including wireless network deployment and maintenance.

Network Engineer, Volkswagen/ HCL, Chattanooga TN

Sept 2020- Nov 2021

- Managed and resolved various network-related tickets, including routing, switching, and wireless connectivity issues, reducing network issues by 70%.
- Configured and managed network routers, switches, ISE, wireless access points, wireless controllers, device configurations, and patch cables (ethernet and fiber) in LAN, WAN and data center environment.
- Monitored networks using SolarWinds for protocol (SIEM) and hardware failures to ensure 100% network availability, designed a network and implemented overall network strategy for a new branch deployment in the campus from 0 to 100% functional.
- Efficiently diagnosed and troubleshot a wide range of infrastructure issues (routing, switching, network Load-balancing, and Wireless) quickly and efficiently, inventory management aligning with business needs.

Infrastructure Engineer, FedEx (Contract), Lakeland FL

Jun 2019- July 2020

- Led the configuration of 800 FedEx sites with new switches on an LCM (life cycle management) project with FedEx using OOB and migrated traffic from the 2K switch to the new 9K series switch and troubleshot.
- Managed different MPLS lines and monitored utilization via tools like NetDB, SolarWinds, SIEM performance monitoring tools to disconnect unnecessary lines. Environment: Cisco Routers (4331, 4351, 4451, 2900), Cisco switches(9300L, 2960, 3850, security switches), troubleshooting BGP over MPLS, iWAN.
- Configured 50 iWAN sites by creating and deploying new configurations with tunnels on the routers using FedEx tools. Planned after-business hours IOS updates for devices.

Volunteer Researcher, Missouri University of Science and Technology

Aug 2018- May 2019

- Research work based on the HWMP protocol in Ad-hoc wireless mesh network behavior, altering the behavior, network testing/ troubleshooting with Raspberry Pi as Ad-hoc devices and Linux tools like tcp dump, iperf3, iw dev.

Project Coordinator- Networking, Ergode IT Services, India

Nov 2015- July 2016

- Coordinated with clients for LAN/WAN topology design, implementation of site-to-site VPN, and route-filters like ACL management, 4900, 2960 switches and routers, configuration and troubleshooting of routing protocols, QoS
- Configured and troubleshot small, and large networks, configured and managed VLANs, VTP, security policies, and voice or video networks; and deployed the IP address space effectively using subnets and NAT.

Technical Support Engineer, Aasma Solutions, India

Feb 2012- Mar 2014

- Configured, tested, and troubleshot network operations, network infrastructure and wireless access on a call with ticket management over the Cisco Nexus environment.
- Routing-related tasks included providing Cisco router configuration and change management and providing technical support for Cisco router configurations and installation for internal and external partners, resolving complex network problems.

EDUCATION

MS in Computer Engineering, Missouri University of Science and Technology, Missouri

Aug 2016- May 2018

BE in Electronics and Telecommunication, University of Mumbai, India

Aug 2007- May 2011

MBA-IT, Campbellsville University, Kentucky,

July 2021-Jan 2024

CERTIFICATIONS

- CISCO certification: CCNA Routing & Switching 2020.
- Prince2 Foundation and Practitioner Certification in Project Management and ITIL 2015.
- AWS Advanced Networking specialty certification (ongoing).